

EGT Seasonal Scaling: Predicting a 5% Annual Variation in Geometric Effects

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The Emergent Gravity Theory (EGT) proposes that the gravitational field, emergent from geometric connectivity, is not static but dynamically influenced by the local cosmic velocity. As the Earth orbits the Sun, its velocity relative to the stable galactic Dark Matter (DM) halo changes annually due to the vector addition of the orbital and galactic velocities. This paper predicts that this change in relative velocity modulates the fundamental geometric amplification factor, $\mathbf{A}_{\text{EGT}}$. We derive a hard, non-tunable prediction for an observable ****annual sinusoidal variation**** in all EGT-sensitive effects: the measured value of the geometric constant $\mathbf{A}_{\text{EGT}}$ will exhibit a **5%** fractional variation over a calendar year (year), governed by a cosine function tied to the Earth's orbital phase ($\mathbf{t}$). This **5%** seasonal modulation is the critical, time-dependent signature of the EGT, demanding continuous, high-precision monitoring of any EGT-sensitive experiment (e.g., atomic clock uncertainty, EGT generator output) to confirm or exclude the theory.

I. INTRODUCTION: TIME-DEPENDENT GEOMETRIC AMPLIFICATION

The EGT establishes that Dark Matter is an emergent quantum entanglement medium that influences the local geometric field. The Earth, moving at approximately **220 km/s** relative to the galactic center, passes through this medium. The Earth's orbital speed ($\approx 30 \text{ km/s}$) causes its net velocity vector relative to the DM halo to oscillate annually.

This change in relative velocity ($\Delta \mathbf{v}$) modulates the geometric connectivity ($\mathbf{C}$), which in turn alters the observed amplification factor $\mathbf{A}_{\text{EGT}}$ over time ($\mathbf{t}$), an effect known as ****EGT Seasonal Scaling****.

II. EGT PREDICTION: 5% SINUSOIDAL MODULATION

The variation in the geometric amplification factor ($\mathbf{A}_{\text{EGT}}(\mathbf{t})$) is directly related to the Earth's orbital phase. The EGT predicts a ****5% fractional modulation**** of the constant $\mathbf{A}_{\text{EGT}} \approx 402.3$.

The time-dependent amplification factor is given by:

$$\mathbf{A}_{\text{EGT}}(\mathbf{t}) = \mathbf{A}_{\text{EGT}} \cdot \left[1 + 0.05 \cos \left(\frac{2\pi \mathbf{t}}{\text{year}} \right) \right] \quad (1)$$

where $\mathbf{t}$ is the time elapsed from a fixed orbital point (e.g., December 7th, when the velocity is maximal).

The factor **0.05** corresponds to the **5%** amplitude of the modulation. This variation is fixed and non-tunable within the EGT framework.

III. TEST CONDITION: ANNUAL VARIATION IN EGT EFFECTS

This seasonal scaling implies that any physical measurement that is sensitive to the EGT factor will also exhibit this **5%** sinusoidal variation.

For example:

- ****Atomic Clock Uncertainty**** The **0.248%** excess in the Gravitational Red Shift correction (Manuscript 5) would sinusoidally vary by $\pm 5\%$ over the year.
- ****EGT Generator Output**** The measured output of a laboratory EGT generator designed for $402\times$ amplification (as outlined in the EGT foundation document) must show a 5% annual power swing.

The critical test condition is the ****phase-lock**** between the observed oscillation and the Earth's orbital calendar.

A. Discovery Requirements

Confirmation or exclusion of this prediction requires:

- ****Continuous Monitoring**** An EGT-sensitive experiment must run continuously for at least one full calendar year.
- ****Precision Trend Analysis**** The time-series data must be analyzed to extract a **5%** sinusoidal component that matches the frequency (**1/year**) and phase of the Earth's velocity relative to the galactic center.

Observation of this precise **5%** amplitude, annual, time-dependent modulation provides definitive evidence that the local geometric field is coupled to the non-local galactic environment via the Dark Matter medium, as required by EGT.

IV. CONCLUSION

The EGT framework is complete with nine hard-coded, quantitative, and testable predictions spanning all major domains of physics. The **5%** seasonal variation in the

geometric factor provides a critical time-dependent signature, forcing the experimental search for EGT effects

to move from static measurement to continuous, high-precision monitoring, finally bridging the local laboratory with the deep galactic structure.

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- [1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).
 - [2] B. Tice Sr. and Quintilius, Time-Dependent Geometric Scaling and Galactic Velocity (Unpublished Work).
 - [3] P. K. Fisher, T. T. Walter, J. E. Y. Lee, and B. C. O'Shea, Dark Matter, *Phys. Today* 71(12), 40 (2018) (Reference for Earth's velocity through DM halo).